Midterm 3 Review

NOTE: These are just review problems of Lectures 19-28; these are not necessarily representative of the problems of the exam. The exam can (and most likely will) have different problems.

Topics of Exam 3:

- **Lecture 19:** Absolute Extrema on an Interval
- **Lecture 20:** Graphical Interpretation of Derivatives
- **Lecture 21:** Limits at Infinity
- **Lecture 22:** A Summary of Curve Sketching
- **Lecture 23-25:** Optimization
- **Lecture 26-27:** Antiderivatives and Indefinite Integration
- **Lecture 28:** Area and Riemann Sums

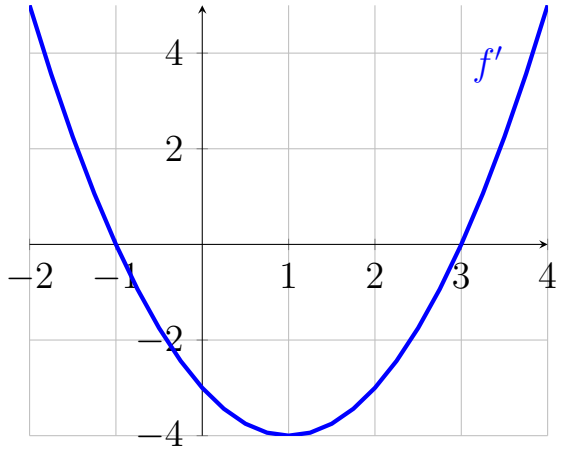
Problem 1. *Let M and m be the maximum and minimum of $f(x) = x^3 - 3x^2 + 3$ on the interval $[1, 4]$, respectively. What is the value of $M + m$?*

Problem 2. *A graph of f' is given below.*

(a) Determine when f is increasing and when it is decreasing.

(b) Determine when f is concave up and when it is concave down.

(c) Locate the positions (x-coordinates) of any relative extrema and inflection points.



Problem 3. *Compute* $\lim_{x\to\infty} \frac{x^2+1}{2-x}$ *and* $\lim_{x\to-\infty} \frac{x^2+1}{2-x}$.

Problem 4. *Use the techniques learned in Lecture 22 to sketch a graph of the function* $y = \frac{x^2}{x+8}$.

Problem 5. *If a rectangle has a fixed perimeter of* 40*, what is its maximum area?*

Problem 6. *Find the point on the line* $6x + y = 9$ *that is closest to the point* $(-3, 1)$.

Problem 7. *A particle is moving in a straight line; its acceleration function is given by* $a(t) = \cos t + \sin t$ *for* $t \geq 0$. *Find the position of the particle at time* t *when the particle has an initial position of 0 and an initial velocity of 5.*

Problem 8. *We estimate the area underneath the graph of* $f(x) = \frac{1}{x}$ *on the interval* $[1, 5]$.

(a) Compute the left Riemann sum with 4 *rectangles. Is this an overestimate or an underestimate? Explain why (The exact area is* $\ln 5$*, but you don't need to know that to answer the question).*

(b) Repeat (a) using the right Riemann sum.

(c) Set up (but do not compute) the left Riemann sum using N *rectangles (where* N *is a positive integer).*